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304 Stainless Steel

Categories: [Metal](#); [Ferrous Metal](#); [Heat Resisting](#); [Stainless Steel](#); [T 300 Series Stainless Steel](#)

Material Notes:

This MatWeb entry has typical values for specimens of unknown form/heat treatment but appear to be annealed samples. We are creating other entries for more specific heat treatments and forms. Austenitic Cr-Ni stainless steel. Better corrosion resistance than Type 302. High ductility, excellent drawing, forming, and spinning properties. Essentially non-magnetic, becomes slightly magnetic when cold worked. Low carbon content means less carbide precipitation in the heat-affected zone during welding and a lower susceptibility to intergranular corrosion.

Applications: beer kegs, bellows, chemical equipment, coal hopper linings, cooking equipment, cooling coils, cryogenic vessels, dairy equipment, evaporators, flatware utensils, feedwater tubing, flexible metal hose, food processing equipment, hospital surgical equipment, hypodermic needles, kitchen sinks, marine equipment and fasteners, nuclear vessels, oil well filter screens, refrigeration equipment, paper industry, pots and pans, pressure vessels, sanitary fittings, valves, shipping drums, spinning, still tubes, textile dyeing equipment, tubing.

Corrosion Resistance: resists most oxidizing acids and salt spray.

UNS S30400; AMS 5501, 5513, 5560, 5565; ASME SA182, SA194 (8), SA213, SA240; ASTM A167, A182, A193, A194

Key Words: aisi304, AISI 304, T304, T 304, SUS304, SS304, 304SS, 304 SS, UNS S30400, AMS 5501, AMS 5513, AMS 5560, AMS 5565, AMS 5566, AMS 5567, AMS 5639, AMS 5697, ASME SA182, ASME SA194 (8), ASME SA213, ASME SA240, ASME SA249, ASME SA312, ASME SA320 (B8), ASME SA358, ASME SA376, ASME SA403, ASME SA409, ASME SA430, ASME SA479, ASME SA688, ASTM A167, ASTM A182, ASTM A193, ASTM A194, ASTM A666, FED QQ-S-763, MILSPEC MIL-S-5059, SAE 30304, DIN 1.4301, X5CrNi189, B.S. 304 S 15, EN 58E, PN 86020 (Poland), OH18N9, ISO 4954 X5CrNi189E, ISO 683/13 11, 18-8

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Physical Properties	Original Value	Comments
Density	8.00 g/cc	
Mechanical Properties	Original Value	Comments
Hardness, Brinell	123	Converted from Rockwell B hardness.
Hardness, Knoop	138	Converted from Rockwell B hardness.
Hardness, Rockwell B	70	
Hardness, Vickers	129	Converted from Rockwell B hardness.
Tensile Strength, Ultimate	505 MPa	
Tensile Strength, Yield	215 MPa	
	@Strain 0.2 %	
Elongation at Break	70 %	in 50 mm

Modulus of Elasticity	193 GPa	
Poissons Ratio	0.29	
Shear Modulus	77.0 GPa	
Izod Impact	150 J	V-Notch
	@Temperature -195 °C	
	150 J	V-Notch
	@Temperature 21.0 °C	
Charpy Impact	325 J	

Electrical Properties	Original Value	Comments
Electrical Resistivity	0.0000720 ohm-cm	
	@Temperature 20.0 °C	
	0.000116 ohm-cm	
	@Temperature 650 °C	
Magnetic Permeability	1.020	at RT

Thermal Properties	Original Value	Comments
CTE, linear	17.3 µm/m-°C	
	@Temperature 0.000 - 100 °C	
	17.8 µm/m-°C	
	@Temperature 0.000 - 315 °C	
	18.7 µm/m-°C	
	@Temperature 0.000 - 650 °C	
Specific Heat Capacity	0.500 J/g-°C	
	@Temperature 0.000 - 100 °C	
Thermal Conductivity	16.2 W/m-K	
	@Temperature 0.000 - 100 °C	
	21.5 W/m-K	
	@Temperature 500 °C	
Melting Point	1400 - 1455 °C	
Solidus	1400 °C	
Liquidus	1455 °C	

Component Elements Properties	Original Value	Comments
Carbon, C	<= 0.080 %	
Chromium, Cr	18 - 20 %	
Iron, Fe	66.345 - 74 %	as balance
Manganese, Mn	<= 2.0 %	
Nickel, Ni	8.0 - 10.5 %	
Phosphorus, P	<= 0.045 %	
Silicon, Si	<= 1.0 %	
Sulfur, S	<= 0.030 %	

[References](#) for this datasheet.

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